Annotated Bibliography

Guest lecture: Allosteric Regulation in Protein Interaction Networks

Prepared by Dr. Dorothy Langston, Langston Lab, Haverstone University. Compiled February 9, 2026 for the spring graduate seminar.

1 Voss, E. and Langston, D. (in revision). Allosteric coupling in multi-protein assemblies. The Ashford Journal of Molecular Biology, AJMB-2025-0894. Our own manuscript. Central to the lecture argument that conformational coupling propagates across interface networks.

2 Kirkland, R. and Langston, D. (2024). Network centrality predicts allosteric hotspots. Used for the methods walkthrough on residue ranking. Postdoc Ryan Kirkland led the analysis.

3 Okonkwo, S. (2025). Molecular dynamics of interface flexibility. Internal preprint. Cite only the published figures; the simulation pipeline is unpublished and stays in the lab.

4 Chen, W. (2025). Qualifying exam literature review on coevolution signals. A clean summary of the coevolution methods I want students to read first.

5 Foundational review on cooperativity, classic model. Anchor for the historical framing at the start of the talk, before moving to network methods.

Note: the Voss manuscript remains unpublished. Share slides referencing AJMB-2025-0894 only with the approved collaborator list until the paper is accepted.